

Dr. Marcello Maugeri

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PROFESSIONAL SUMMARY

Security Research Engineer with a PhD in Computer Science and expertise in cybersecurity and automated security testing. Designs and builds tools for vulnerability discovery, resulting in peer-reviewed publications and open-source software. As AI capabilities advance, his research is expanding into AI-assisted security testing, including the evaluation of AI systems for security and safety.

PROFESSIONAL AND RESEARCH EXPERIENCE

Incoming M3 Fellow, AI Security

Sep 2026 – Oct 2026

M3 Fellowship, London, United Kingdom

- Selected as one of six fellows in the inaugural M3 cohort, an AI security research fellowship for senior security professionals run by Constellation, Kairos and AI Security Bootcamp (AISB). Will develop a self-directed project addressing an open problem in AI security, with peer review and feedback from field experts.

Founding Member & Senior Cybersecurity Research Engineer

Sep 2025 – Jul 2026

CybPass Ltd., Sheffield, United Kingdom (Remote)

- Led development of a proof-of-concept for an AI management product based on ISO/IEC 42001. The PoC demonstrated the product's technical feasibility and supported the startup's early-stage development ahead of its spin-out from The University of Sheffield. Translated ISO/SAE 21434 clauses covering Threat Analysis and Risk Assessment (TARA) into technical requirements for an automotive security product.
- Led successful applications to DRIVE35 Mobilise and Cyber Runway Grow, a government-backed accelerator delivered by Plexal. The Mobilise project secured up to £173,500 to develop a tool for assessing the safety and security of AI systems and producing evidence for compliance with industry requirements.
- Designed and developed a MetaDrive-based benchmark for training Reinforcement Learning attackers and evaluating adversarial manoeuvre attacks against autonomous-driving policies. Published the paper at USENIX VehicleSec'26 and released the benchmark as open-source software at <https://github.com/CybPass-AI/AMAB>.
- Conducted security evaluations of Vision-Language Models (VLMs) and Vision-Language-Action (VLA) models for autonomous driving, including NVIDIA Cosmos and Alpamayo models, against state-of-the-art adversarial visual perturbations and backdoor attacks.

Doctoral Researcher, Software Security

Oct 2021 – Feb 2026

University of Catania, Catania, Italy

- Formalised the fork-awareness property to characterise a fundamental limitation of coverage-guided fuzzing when the software under test (SUT) comprises multiple processes. Identified the limitation through an evaluation of 14 fuzzers.
- Designed ForkFuzz, a fork-aware coverage-guided fuzzer built in C on top of Honggfuzz. It uses the Linux ptrace system call to monitor the process tree.
- Co-designed LibAFLstar, a stateful protocol fuzzer built with Rust and LibAFL, and led target integration and experimental evaluation across six FTP, RTSP and HTTP implementations. Built the evaluation infrastructure with Docker and reconstructed the RTSP and HTTP state models through RFC analysis and network traffic captured with Wireshark. LibAFLstar ran over 30× faster and achieved 1.4× more coverage than AFLNet and ChatAFL on average.
- Designed BenGQL, a benchmarking platform built with Bash and Docker for evaluating automated testing approaches across GraphQL applications. Designed KrakQL, a multi-agent tool built with Google ADK for blind GraphQL schema introspection. Across seven applications, KrakQL improved schema coverage by 140% while using 134× fewer HTTP requests than the state-of-the-art baseline.

AI & Data Engineer (Contract)

Oct 2025 – Jan 2026

University of Catania, Department of Political and Social Sciences, Catania, Italy

- Developed a Python document-processing pipeline using Mistral OCR and LangGraph-based LLM agents to extract and structure multilingual parliamentary questions and answers from historical scans of the Official Journal of the European Union, including French-only editions dating back to the 1950s.
- Curated a dataset of more than 80,000 structured parliamentary questions for EUQuest, a PRIN 2022 project funded through the EU's NextGenerationEU programme (CUP E53D23006720006), combining automated processing with targeted manual correction of OCR and extraction errors.

AI Agent Researcher (Contract)

Nov 2024 – Mar 2025

RNDM, Manchester, United Kingdom (Remote)

- Led the design and development of a LangGraph-based multi-agent system for DeFi lending-protocol analytics. Built the supporting data pipeline and Neo4j knowledge graph for GraphRAG retrieval.
- Evaluated LLM-based time-series forecasting against established statistical models for DeFi interest-rate optimisation and found no reliable performance advantage.

Research Engineer, PECS Project

Jan 2024 – Jun 2024

University of Catania, Catania, Italy

- Developed the successful team proposal for the PECS project as part of the nas.inf research group, securing €117,000 through NGI TrustChain Open Call #2.
- Reviewed Federated Learning (FL) approaches for automotive applications, then designed and implemented a user-empowered FL prototype using Flower, Python, Kotlin, Android Automotive and TensorFlow Lite, allowing users to opt out of distributed model training.
- Evaluated the prototype on an engine-fault classification case study using EngineFaultDB. Published the resulting paper at IEEE Blockchain 2024 and released the implementation as open-source software at <https://github.com/marcellomaugeri/User-Empowered-Federated-Learning-in-Automotive>.

Visiting PhD Student

Oct 2023 – Mar 2024

King's College London, London, United Kingdom

- Conducted research under the supervision of Prof. Fabio Pierazzi on black-box security testing for GraphQL APIs using Deep Reinforcement Learning and fuzzing. Co-designed and developed Wendigo, a black-box approach that uses Proximal Policy Optimisation (PPO) and the target GraphQL schema to discover Denial-of-Service queries.
- Evaluated Wendigo against EvoMaster, a state-of-the-art fuzzing tool for GraphQL APIs. Published the paper at the IEEE Workshop on Deep Learning Security and Privacy (DLSP), co-located with IEEE S&P 2024, and released the implementation as open-source software at <https://github.com/isneslab/Wendigo>.

Software Engineer & Technical Project Manager

Mar 2023 – Nov 2024

Helpcode, Genoa, Italy (Remote)

- Conceived, developed and launched a Flutter educational platform for The Water Code, enabling students aged 10 to 14 to apply computational thinking to environmental challenges. Built the backend using Firebase Authentication, Cloud Functions, Cloud Storage and a managed Firebase database: <https://app.thewatercode.it/>.
- Wrote the concept and proposal for the platform component of The Water Code, a €699,933 AICS-funded project delivered by a consortium of 10 organisations. Managed the development work and coordinated the graphic designer and storyteller using Jira.

Cybersecurity Programme Coordinator & Instructor

Nov 2021 – Jun 2023

CINI Cybersecurity Lab UNICT Hub, Catania, Italy

- Coordinated the University of Catania hub of CyberChallenge.IT, Italy's national cybersecurity training and Capture-the-Flag (CTF) programme. Managed communications with central organisers, 5–7 tutors, classroom logistics and programme delivery for 25 participants per annual cohort.
- Taught practical modules in Cryptography and Network Security, covering attacks against symmetric and asymmetric cryptographic schemes, network traffic and protocol analysis with Wireshark, and deployment of Docker containers for the local CTF infrastructure.

Research Engineer (Internship)

May 2021 – Aug 2021

Robert Bosch GmbH, Renningen, Germany

- Developed a coverage-guided fuzzing harness for shared-memory IPC, integrating Honggfuzz with QEMU user-mode plugins that intercepted system calls associated with POSIX and System V shared memory.
- Applied the harness to Eclipse iceoryx and uncovered undisclosed bugs in RouDi, its central daemon for shared-memory management and service discovery. Passed the findings to the development team for triage and documented the work in the MSc dissertation "Shared Memory Fuzzing".

Founding Member & Mobile Software Engineer

May 2020 – Apr 2021

Codedix, Catania, Italy

- Developed and released three native and cross-platform mobile applications for iOS and Android using Swift, Objective-C, Kotlin, Flutter and Dart. Applied MVC architecture and integrated Firebase services (Auth, Firestore, Cloud Storage).
- Worked directly with clients to translate requirements into production releases and helped build Codedix's initial mobile application portfolio before the company was incorporated.

Research Intern

Oct 2020 – Dec 2020

Robert Bosch GmbH, Renningen, Germany (Remote)

- Conducted a systematic review of 42 embedded-systems fuzzing papers and their associated tools, comparing techniques that tested software on physical hardware, emulated its execution environment, or abstracted hardware interactions.
- Analysed selected techniques, documented their strengths and limitations, and contributed to the comparative table and identification of research gaps. Co-authored the resulting review, published in the Q1 Springer Nature journal Cybersecurity.

EDUCATION

PhD in Computer Science University of Catania, Italy. Dissertation Title: <i>Automation Challenges and Solutions in Coverage-Guided Fuzzing of Multiprocess Software Systems.</i>	Oct 2021 – Sep 2025
Erasmus+ Exchange Programme University College Dublin, Ireland Completed the following modules as an Erasmus+ exchange student: Deep Learning (COMP47650), Secure Software Engineering (COMP47910), Data Science in Python (COMP41680), Mobile App Dev - Cocoa Touch (COMP47390), and Advances in Wireless Networking (COMP40660).	Oct 2019 – Sep 2021
MSc Computer Science (Major: Cybersecurity and Distributed Systems) University of Catania, Italy Thesis Title: <i>Shared Memory Fuzzing.</i> (Grade 110/110 with honours)	Oct 2019 – Sep 2021
BSc Computer Science (Major: Networks and Cybersecurity) University of Catania, Italy Thesis Title: <i>3DCP Dynamic Domotic Device Configuration Protocol and Smart Hub.</i> (Grade: 110/110 with honours)	Oct 2016 – Nov 2019

SELECTED PUBLICATIONS

- Emamy, A., Maugeri, M., Lin, P., & Pasikhani, A. (2026). **WIP: A Benchmark for Security Evaluation of Adversarial Manoeuvre Attacks in Autonomous Driving.** *Usenix VehicleSec'26.*
- Maugeri, M., Angamo, A., & Bella, G. (2025). **KrakQL: LLM-Guided Blind Introspection of GraphQL Schemas.** *Symposium on Search-Based Software Engineering (SSBSE).* doi:10.1007/978-3-032-24839-8_2.
- Angamo, A., & Maugeri, M. (2025). **BenGQL: An Extensible Benchmarking Framework for Automated GraphQL Testing.** *International Conference on Automated Software Engineering (ASE).* doi:10.1109/ASE63991.2025.00376.
- Maugeri, M., Castiglione, G., Raciti, M., & Bella, G. (2025). **AI-related Vulnerabilities within CVEs: Are We Ready Yet? A Study of Vulnerability Disclosure in AI Products.** Workshop on Artificial Intelligence and Security (AISec). doi:10.1145/3733799.3762969.
- Daniele, C., Bethe, T., Maugeri, M., Continella, A., & Poll, E. (2025). **LibAFLstar: Fast and State-aware Protocol Fuzzing.** *European Symposium on Research in Computer Security (ESORICS).* doi:10.1007/978-3-032-07894-0_6.
- McFadden, S., Maugeri, M., Hicks, C., Mavroudis, V., & Pierazzi, F. (2024). **WENDIGO: Deep Reinforcement Learning for Denial-of-Service Query Discovery in GraphQL.** *IEEE Symposium on Security and Privacy Workshops (DLSP - Deep Learning Security and Privacy Workshop).* doi:10.1109/SPW63631.2024.00012.
- Maugeri, M., Daniele, C., & Bella, G. (2024). **Forkfuzz: Leveraging the Fork-Awareness in Coverage-Guided Fuzzing.** *ESORICS 2023 Workshop (SecAssure).* doi:10.1007/978-3-031-54129-2_17
- Eisele, M., Maugeri, M., Shriwas, R., Huth, C., & Bella, G. (2022). **Embedded fuzzing: a review of challenges, tools, and solutions.** *Springer Cybersecurity.* doi:10.1186/s42400-022-00123-y.

HONOURS AND AWARDS

• 3rd place, EulerSwap Builder Competition (5,000\$)	Jul 2025
• Honourable Mention, Vibe Coding Hack by Encode - Polygon Track (200\$)	Jun 2025
• 2nd place, Encode AI London 2025 Hackathon – Starkware AI x DeFi Track (300\$)	Apr 2025
• 2 awards at ETHOxford Hackathon - Flare's track 1: pool prize - NERDo Awards (2,000\$)	Feb 2025
• 1st place, Encode London 2024 Hackathon - RNDM Agent Track. (5,000\$)	Oct 2024
• 1st place tied (2 nd place by score time) of CyberChallenge.IT final CTF in Catania.	Jun 2021

CERTIFICATIONS

• Google Generative AI Leader Certification	Aug 2026
• Google Professional Cloud Architect Certification	Feb 2026
• IEEE Software Professional Certificate Level 2	Nov 2025
• ISC2 Certified in Cybersecurity (CC)	Oct 2024
• Google Project Management Specialization	Dec 2023
• Google Cybersecurity Specialization	Dec 2023